Solution, simulation and analysis of the angular dynamics of a free body with a magnetic moment subject to a rotating magnetic field

▾ Abstract/Research Context

This application is a symbolic, numerical, and visual analysis framework for studying the angular dynamics of free or magnetically bounded bodies subjected to rotating magnetic fields. Magnetic bounding by rotating fields is a relatively new phenomenon in which a rigid body with a magnetic moment can be trapped at a precise location by a rotating magnetic field with a gradient, without requiring a local minimum of magnetic potential. The effect is essentially realizable with dipole magnets. In this interaction, the body couples to the field through its magnetic moment, which is fixed in the body frame, and may undergo translational and angular oscillations of typically small amplitude. These oscillations allow the body to find a quasi-stable equilibrium with the driving field and the confinement mechanism which impossible within magnetostatic except using diamagnetism.

This work focuses on the angular part of the dynamics, which is responsible for keeping the body in proper alignment with the rotating field. In a basic model, the angular oscillation consists of a symmetric conical motion of the body's magnetic moment around the axis of the rotating field syncronized with this field. This alignment mechanism can be associated with the phase-lag property of driven oscillators; in this case the phase is held close to π, causing the body's magnetic moment to be antiparallel to the field in the azimuthal components, while parallel alignment occurs in the zenithal components of the magnetic moment and the rotating field.

Since antiparallel alignment results in repulsion and parallel in attraction, the stability in translational degrees of freedom is obtained by the system ability to adjust these opposing effects by variation of the zenithal angle of the body -- which, in turn, corresponds to the amplitude of this driven oscillation as a function of the strength of magnetic torque acting on the body.

As the body moves closer in the field gradient, the stronger field increases the magnetic torque, causing the oscillation amplitude to increase. This corresponds to a larger zenithal angle and makes the antiparallel alignment more pronounced. Consequently, the repulsive component grows faster than the attractive component, producing an effective positive translational stiffness.

A related self-regulating mechanism can also arise from translational oscillations, especially in the radial plane with respect to the rotation axis of the driving field in small circles syncronized by the rotating field. Through the phase-lag response of the driven motion, the displacement of the dipole can become nearly opposite in phase to the effective attractive force. This shifts the dipole (therefore the body) in the field gradient in a way that favors the repulsive component of the magnetic interaction, so that the attractive force is effectively used against itself. The result is an effective positive translational stiffness. Although this displacement may be small, it can still be sufficient to produce a bound state in cases where angular oscillations are constrainted or suppressed.

In the model implemented here, the angular degrees of freedom are described by the three Euler angles θ, ϕ, and ψ. The angle ψ denotes rotation of the body about its magnetic axis, while ϕ and θ describe the azimuthal and zenithal orientation of the magnetic moment in a coordinate system whose z-axis is the rotation axis of the driving field. The magnetic moment is assumed to be fixed in the body frame, while the applied magnetic field rotates with user-defined angular velocity, field inclination, and strength.

The equations of motion are derived using the Euler–Lagrange formalism in the Euler-angle coordinates (θ, ϕ, ψ), for a rigid body with a diagonal moment-of-inertia tensor whose one principal axis coincides with the magnetic axis. Axisymmetric and isotropic cases are also covered as reductions of the general model.

For an axisymmetric body, the equations of motion admit synchronized steady-state solutions in which the vectors entering the dynamics remain time-invariant in the reference frame of the rotating magnetic field. In this rotating-field frame, the explicit time dependence cancels and the equations of motion become autonomous. These equilibria, supported by numerical simulations and experimental observations, provide the operating points for local linearization. The equations of motion are then expanded to first order about the equilibrium state defined by the generalized coordinates and their first derivatives, yielding the state-matrix representation used for eigenvalue analysis.

The eigenvalue analysis exposes the local stability characteristics and natural angular frequencies of the synchronized solution. In the absence of damping, the system can be marginally stable, with eigenvalues whose real parts are zero. In the presence of damping, stable solutions become asymptotically stable when the real parts of all eigenvalues are negative. When the field inclination angle γ is reduced below the required stability threshold, one real part becomes positive, clearly indicating an unstable regime.

The simulations and eigenvalue results support each other in identifying stability limits and reproducing accurately the observed eigenfrequencies. The imaginary parts of the eigenvalues give the natural angular frequencies of the synchronized solution. In the stable regime, two principal eigenfrequencies are typically observed close to the driving frequency; for small oscillation amplitudes, one lies slightly below the driving frequency and the other slightly above it. The separation of these frequency branches is mainly controlled by the field inclination angle γ, while increasing oscillation amplitude tends to shift both frequencies downward. This downward shift can be interpreted as a softening-type nonlinear effect. Through parameter sweeps, the utility makes it possible to observe frequency branching, stability boundaries, damping effects, and the influence of inertia ratios.

Simulations further indicate the existence of excited bounded modes within the stable regime. In these modes, the zenithal angle of the magnetic moment can vary over a large range, and the corresponding state-space trajectories may form cycloidal, rosette-like, looped, or otherwise structured patterns. In damped cases, these trajectories can converge toward attractors or preferred bounded regions in state space, indicating stable excited forms of motion in addition to the basic synchronized equilibrium. The built-in state-variable plots, phase portraits, and polar plots provide a practical way to examine these modes and compare them with regular synchronized motion, beat formation, nonlinear amplitude limitation, and the onset of instability.

Section C analyzes the behavior of the state variables in the presence of damping. In particular, it is shown that, in the absence of electromagnetically induced spin-up torques, ψ converges to −ω cos θ. This condition corresponds to a zero z-component of the body's angular velocity in local coordinates and provides a useful reference state for setting up damped simulations.

Direct nonlinear simulations and eigenvalue results support each other in identifying stability limits and reproducing the observed eigenfrequencies. Simulations also reveal an angular instability mechanism when the body has a small non-axisymmetry, which is often encountered in experiments. In such cases, the nonlinear dynamics of the system can use this asymmetry to promote resonant angular modes.

Section E derives Ucar (2021)'s semi-empirical stability criterion from eigenvalue bifurcation and reformulates the formula in terms of the driving-field elevation angle γ. The formula is applicable to isotropic bodies under the condition ψ = −ω under zero damping and is found to predict stability limits in agreement with both eigenvalue analyses and direct simulations.

A central purpose of the application is to support the investigation of magnetic bound-state dynamics beyond simple numerical simulation. The symbolic derivation of the equations of motion allows the assumptions, coordinate definitions, and model reductions to be inspected directly. The numerical solver enables time-domain simulations, while the animation, plotting, and phase-portrait tools help visualize the physical motion and the evolution of state variables.

The application also includes linearization and eigenvalue analysis tools in Section D. These are useful for studying local stability around synchronized operating states, identifying natural angular frequencies, and comparing predicted small-signal behavior with frequencies observed in full nonlinear simulations. By sweeping parameters such as field rotation rate, field inclination, damping, magnetic torque strength, inertia ratios, and initial conditions, the user can observe how stable, unstable, quasi-synchronized, and transitional regimes emerge.

Therefore, this utility can be used as both a research notebook and an exploratory computational laboratory for angular magnetic dynamics. It may be useful for studying magnetically bounded motion, rotating-field stabilization, resonance-like behavior, damping-dependent equilibrium states, non-axisymmetric instability mechanisms, and the relationship between nonlinear simulations and linearized eigenvalue spectra.

▸ Quick Guide

▾ A. Derivation of Equations of Motion

Equations of angular motion of a body subject a dynamic magnetic torque are calculated in body coordinates as follow.
Equations generated by this flow are used in numeric integration solver in order generate simulation data.

1. Symbols and Generalized Coordinates

$$\omega,\, I,\, mB,\, \gamma,\, \xi,\, t \in \mathbb{R},\quad \theta(t),\ \phi(t),\ \psi(t),\ \dot\theta = \frac{d\theta}{dt},\ \dot\phi = \frac{d\phi}{dt},\ \dot\psi = \frac{d\psi}{dt}.$$

Here ω is the angular velocity of the rotating field, I the diagonal moment-of-inertia tensor with components (I_1, I_2, I_3). mB is a scalar corrseponding product of strengths of magnetic moment $\vec m$ of the body and strength of the rotating field $\vec B$. γ denotes the elevation angle the vector $\vec B$ makes with its rotation plane (xy), also called tilt angle, ξ the damping coefficient and t the time. $\theta(t)$, $\phi(t)$, $\psi(t)$ are the rotations of the body defined in Euler rotation matrix in ZXZ scheme.

2. ZXZ Rotation Matrix

$$R = R_z(\phi)\, R_x(\theta)\, R_z(\psi),\quad R_z(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix},\quad R_x(\beta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\beta & -\sin\beta \\ 0 & \sin\beta & \cos\beta \end{pmatrix}.$$

3. Angular Velocity Vector in Body Frame

$$[\nu]_\times = R^T \dot R \implies \nu = \mathrm{vee}(R^T \dot R) = \begin{pmatrix} (R^T \dot R)_{3,2} \\ (R^T \dot R)_{1,3} \\ (R^T \dot R)_{2,1} \end{pmatrix} = \begin{pmatrix} \dot\theta\cos\psi + \dot\phi\sin\psi\,\sin\theta \\ -\dot\theta\sin\psi + \dot\phi\cos\psi\,\sin\theta \\ \dot\phi\cos\theta + \dot\psi \end{pmatrix}.$$

4. Rayleigh Dissipation Function

$$\mathcal{R} = \tfrac12 \xi\, \|\nu\|^2 = \tfrac12 \xi \left(2\cos(\theta)\dot\phi\dot\psi + \dot\phi^2 + \dot\psi^2 + \dot\theta^2\right).$$

5. Unit Vector of Rotating Magnetic Field

$$\hat B = \begin{pmatrix} \sin\omega t\, \cos\gamma \\ -\cos\omega t\, \cos\gamma \\ \sin\gamma \end{pmatrix}.$$

6. Unit Vector of Body's Magnetic Moment

$$\hat m = R \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \sin\phi\,\sin\theta \\ -\cos\phi\,\sin\theta \\ \cos\theta \end{pmatrix}.$$

7. Moment of Inertia Tensor For This Model

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}.$$

8. Kinetic and Potential Energy

$$T = \tfrac12 \nu I \nu = \frac{I_1\left(\sin\psi\,\sin\theta\,\dot\phi + \cos\psi\,\dot\theta\right)^2}{2} + \frac{I_2\left(\sin\psi\,\dot\theta - \sin\theta\,\cos\psi\,\dot\phi\right)^2}{2} + \frac{I_3\left(\cos\theta\,\dot\phi + \dot\psi\right)^2}{2},$$

$$V = -mB\,(\hat m\cdot\hat B) = -mB\,(\sin\gamma\,\cos\theta + \sin\theta\,\cos\gamma\,\cos(\omega t - \phi)).$$

Kinetic Energy of a Body with Isotropic Moment of Inertia

$$T_{iso} = \frac{I\left(2\cos\theta\,\dot\phi\dot\psi + \dot\phi^2 + \dot\psi^2 + \dot\theta^2\right)}{2}.$$

9. Euler–Lagrange Equations with Damping

$$\frac{d}{dt}\left(\frac{\partial(T-V)}{\partial \dot q_i}\right) - \frac{\partial(T-V)}{\partial q_i} = Q_i^{(\mathrm{diss})},\quad Q_i^{(\mathrm{diss})} = -\frac{\partial\mathcal R}{\partial \dot q_i},\quad q_i \in \{\theta, \phi, \psi\}.$$

10. Solve for the Angular Accelerations

$$\ddot\theta \to \theta_{dd},\quad \ddot\phi \to \phi_{dd},\quad \ddot\psi \to \psi_{dd},$$

$$\begin{cases} E_\theta(\theta, \phi, \psi, \dot{}, \ddot{}) = 0, \\ E_\phi(\ldots) = 0, \\ E_\psi(\ldots) = 0 \end{cases} \implies \{\theta_{dd} = f_1(\cdots),\ \phi_{dd} = f_2(\cdots),\ \psi_{dd} = f_3(\cdots)\}.$$

▾ B. Basic Definitions and Equations of Motion

1. General rotation matrix in ZXZ convention based on Euler angles θ, ϕ, and ψ

$$R = R_z(\phi)\, R_x(\theta)\, R_z(\psi) = \begin{pmatrix} -\sin\phi\,\sin\psi\,\cos\theta + \cos\phi\,\cos\psi & -\sin\phi\,\cos\psi\,\cos\theta - \sin\psi\,\cos\phi & \sin\phi\,\sin\theta \\ \sin\phi\,\cos\psi + \sin\psi\,\cos\phi\,\cos\theta & -\sin\phi\,\sin\psi + \cos\phi\,\cos\psi\,\cos\theta & -\sin\theta\,\cos\phi \\ \sin\psi\,\sin\theta & \sin\theta\,\cos\psi & \cos\theta \end{pmatrix}.$$

2. Unit vectors $(\hat B, \hat m)$ of the rotating magnetic field and the body's magnetic moment

$$\hat B = \begin{pmatrix} \sin\omega t\,\cos\gamma \\ -\cos\omega t\,\cos\gamma \\ \sin\gamma \end{pmatrix},\qquad \hat m = \begin{pmatrix} \sin\phi\,\sin\theta \\ -\cos\phi\,\sin\theta \\ \cos\theta \end{pmatrix}.$$

Here ω denotes the angular velocity of $\hat B$ around z-axis, and γ the elevation angle of $\hat B$ from the xy-plane.
The angle ψ does not enter $\hat m$ because it corresponds to a self-rotation around the axis of $\hat m$.

3. Body's angular velocity vector ν in the lab and in the body's frames

$$\nu_{body} = \begin{pmatrix} \dot\theta\cos\psi + \dot\phi\sin\psi\,\sin\theta \\ -\dot\theta\sin\psi + \dot\phi\cos\psi\,\sin\theta \\ \dot\phi\cos\theta + \dot\psi \end{pmatrix},\qquad \nu_{lab} = \begin{pmatrix} \dot\theta\cos\phi + \dot\psi\sin\phi\,\sin\theta \\ \dot\theta\sin\phi - \dot\psi\cos\phi\,\sin\theta \\ \dot\psi\cos\theta + \dot\phi \end{pmatrix}.$$

4. Equations of motion based on Euler angles θ, ϕ and ψ, where I_1, I_3 are radial and axial components of the moment-of-inertia tensor of an axisymmetric body, and ξ is the damping coefficient

$$I_1\ddot\theta = \frac{mB\left(-I_2\sin\gamma\,\sin\theta + (-I_1 + I_2)\,(\sin\gamma\,\sin^2\psi\,\sin\theta + \sin(\omega t - \phi)\sin\psi\,\cos\gamma\,\cos\psi) + (I_1\sin^2\psi + I_2\cos^2\psi)\cos\gamma\,\cos(\omega t - \phi)\cos\theta\right) + \xi\left(-I_2\dot\theta + (-I_1 + I_2\right.}{}$$

$$I_1\ddot\phi = \frac{mB\left((I_1 - I_2)\sin\gamma\,\sin\psi\,\sin\theta\,\cos\psi + (-(I_1 - I_2)\sin\psi\,\cos(\omega t - \phi)\cos\psi\,\cos\theta + (I_1 - (I_1 - I_2)\sin^2\psi)\sin(\omega t - \phi))\cos\gamma\right) + \xi\left((I_1 - I_2)\sin\psi\,\cos\psi\,\dot\theta + \right.}{}$$

$$I_1\ddot\psi = \frac{{I_1}^2 I_2\left(\left(-\sin^2\theta\,\cos\psi\,\dot\phi^2 + \cos\psi\,\dot\theta^2\right)\sin^5\psi + \sin^6\psi\,\sin\theta\,\dot\phi\dot\theta - 3\sin^4\psi\,\sin\theta\,\dot\phi\dot\theta + 2\sin^3\psi\,\sin^2\theta\,\cos\psi\,\dot\phi^2 - 2\sin^3\psi\,\cos\psi\,\dot\theta^2 + \sin^2\psi\,\sin\theta\,\dot\phi\dot\theta + \sin\psi\,\sin^2\right.}{}$$

5. Reduced equations of motion for an axisymmetric body where $I_1 = I_2$.

$$I_1\ddot\theta = mB\,(-\sin\gamma\,\sin\theta + \cos\gamma\,\cos(\omega t - \phi)\cos\theta) - \xi\dot\theta - \left(I_1\dot\psi + (-I_1 + I_3)\left(\cos\theta\,\dot\phi + \dot\psi\right)\right)\sin\theta\,\dot\phi,$$

$$I_1\ddot\phi = \frac{I_3\dot\psi\dot\theta + mB\sin(\omega t - \phi)\cos\gamma - \xi\sin\theta\,\dot\phi - (2I_1 - I_3)\cos\theta\,\dot\phi\dot\theta}{\sin\theta},$$

$$I_1\ddot\psi = -\frac{I_3\cos\theta\,\dot\psi\dot\theta}{\sin\theta} - \frac{mB\sin(\omega t - \phi)\cos\gamma\,\cos\theta}{\sin\theta} + \xi\left(-\frac{I_1\left(\cos\theta\,\dot\phi + \dot\psi\right)}{I_3} + \cos\theta\,\dot\phi\right) + \left(-I_1\sin\theta + \frac{2I_1}{\sin\theta} + I_3\sin\theta - \frac{I_3}{\sin\theta}\right)\dot\phi\dot\theta.$$

6. Reduced equations of motion for a body with isotropic moment-of-inertia where $I = I_1 = I_2 = I_3$.

$$I\ddot\theta = -I\sin\theta\,\dot\phi\dot\psi + mB\,(-\sin\gamma\,\sin\theta + \cos\gamma\,\cos(\omega t - \phi)\cos\theta) - \xi\dot\theta,$$

$$I\ddot\phi = \frac{I\left(-\cos\theta\,\dot\phi + \dot\psi\right)\dot\theta + mB\sin(\omega t - \phi)\cos\gamma - \xi\sin\theta\,\dot\phi}{\sin\theta},$$

$$I\ddot\psi = \frac{-I\left(\cos\theta\,\dot\psi - \dot\phi\right)\dot\theta - mB\sin(\omega t - \phi)\cos\gamma\,\cos\theta - \xi\sin\theta\,\dot\psi}{\sin\theta}.$$

▾ C. Evaluating State Variables in the Presence of Damping

1. Finding the equilibrium value of $\dot\psi$ under damping ($\xi \neq 0$) for an axisymmetric body

A floating body in the presence of damping, may experience a drag torque that acts on the rotation (spin) about the symmetry (magnetic) axis and can stop that spin if present. This spin corresponds to the z-component of angular velocity ν_z expressed in the body's coordinates. In this section, the equilibrium value of state variable $\dot\psi$ is derived from the equations of motion for an axisymmetric body. It should be noted that a body having zero ν_z in the body's coordinates can still rotate in lab frame which can be seen by tracking a visible marking on the body and measured by a tachometer. This rotation vanishes when $\dot\phi = \omega$ and $\dot\psi = -\omega$, also when $\dot\phi \neq \omega$ and $\dot\psi \approx -\dot\phi$ with a small adjustment.

Equation of motion about $\ddot\theta$:

$$(1)\quad I_1\ddot\theta = mB\,(-\sin\gamma\,\sin\theta + \cos\gamma\,\cos(\omega t - \phi)\cos\theta) - \xi\dot\theta - \left(I_1\dot\psi + (-I_1 + I_3)\left(\cos\theta\,\dot\phi + \dot\psi\right)\right)\sin\theta\,\dot\phi.$$

Equation of motion about $\ddot\phi$:

$$(2)\quad I_1\ddot\phi = \frac{I_3\dot\psi\dot\theta + mB\sin(\omega t - \phi)\cos\gamma - \xi\sin\theta\,\dot\phi - (2I_1 - I_3)\cos\theta\,\dot\phi\dot\theta}{\sin\theta}.$$

Equation of motion about $\ddot\psi$:

$$(3)\quad I_1\ddot\psi = -\frac{I_3\cos\theta\,\dot\psi\dot\theta}{\sin\theta} - \frac{mB\sin(\omega t - \phi)\cos\gamma\,\cos\theta}{\sin\theta} + \xi\left(-\frac{I_1\left(\cos\theta\,\dot\phi + \dot\psi\right)}{I_3} + \cos\theta\,\dot\phi\right) + \left(-I_1\sin\theta + \frac{2I_1}{\sin\theta} + I_3\sin\theta - \frac{I_3}{\sin\theta}\right)\dot\phi\dot\theta.$$

Equilibrium condition admits $\ddot\theta = 0,\ \ \ddot\phi = 0,\ \ \ddot\psi = 0,\ \ \dot\theta = 0$

Equation of motion (2) of $\ddot\phi$ under equilibrium condition

$$(4)\quad 0 = \frac{mB\sin(\omega t - \phi)\cos\gamma - \xi\sin\theta\,\dot\phi}{\sin\theta}.$$

This yields:

$$(5)\quad mB\sin(\omega t - \phi)\cos\gamma = \xi\sin\theta\,\dot\phi.$$

Equation of motion (3) of $\ddot\psi$ under equilibrium condition

$$(6)\quad 0 = -\frac{mB\sin(\omega t - \phi)\cos\gamma\,\cos\theta}{\sin\theta} + \xi\left(-\frac{I_1\left(\cos\theta\,\dot\phi + \dot\psi\right)}{I_3} + \cos\theta\,\dot\phi\right).$$

Substitute (5) in (6)

$$(7)\quad 0 = \xi\left(-\frac{I_1\left(\cos\theta\,\dot\phi + \dot\psi\right)}{I_3} + \cos\theta\,\dot\phi\right) - \xi\cos\theta\,\dot\phi.$$

Simplify

$$(8)\quad \frac{I_1\xi\left(\cos\theta\,\dot\phi + \dot\psi\right)}{I_3} = 0.$$

Eq.8 is satisfied when $\dot\psi = -\cos(\theta)\dot\phi$.

Note: Any initial velocity $\dot\phi$ converges to $-\cos(\theta)\dot\phi$ in the presence of damping.

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2. Finding the equilibrium value of the phase of θ rotation under damping ($\xi \neq 0$) for an axisymmetric body

Equation of motion about $\ddot\theta$ for an axisymmetric body by substitution of $\dot\psi$ by $-\dot\phi\cos\theta$ reads

$$(9)\quad I_1\ddot\theta = I_1\sin\theta\,\cos\theta\,\dot\phi^2 + mB\,(-\sin\gamma\,\sin\theta + \cos\gamma\,\cos(\omega t - \phi)\cos\theta) - \xi\dot\theta.$$

Note that the axial component of moment-of-inertia (I_3) vanishes from the equation of motion. This can be expected since z component of angular velocity vector ν_z in body coordinates becomes zero when $\dot\psi = -\dot\phi\cos\theta$.

By applying equilibrium condition ($\ddot\theta = \dot\theta = 0$) this equation becomes

$$(10)\quad 0 = I_1\sin\theta\,\cos\theta\,\dot\phi^2 + mB\,(-\sin\gamma\,\sin\theta + \cos\gamma\,\cos(\omega t - \phi)\cos\theta).$$

The equation can be satisfied only when the term $\cos\theta\,\cos\gamma\,\cos(\omega t - \phi)$ becomes constant.

Since we are not interested in the cases $\theta = \pi/2$ or $\gamma = \pi/2$, remaining solution is

$$(11)\quad \phi = \omega t + \phi_0.$$

$$(12)\quad 0 = I_1\omega^2\sin\theta\,\cos\theta + mB\,(-\sin\gamma\,\sin\theta + \cos\gamma\,\cos\phi_0\,\cos\theta).$$

The phase of ϕ defined as ϕ_0 can be derived from (5) by substituting (11) into.

$$(13)\quad mB\sin\phi_0\,\cos\gamma = -\omega\xi\sin\theta,.$$

$$(14)\quad \sin\phi_0 = -\frac{\omega\xi\sin\theta}{mB\cos\gamma},.$$

Since $\sin(\phi_0)$ is negative, the angle is equal to $\pi - \phi_0$

$$(15)\quad \phi_0 = \mathrm{asin}\left(\frac{\omega\xi\sin\theta}{mB\cos\gamma}\right) + \pi.$$

Calculate ϕ_0 with current parameters

▾ D. Linearization and Eigenvalue Analysis

This box covers derivation of linearized equations of motion (Jacobians, state matrix) and eigenvalues. This allows one to obtain eigenvalue spectrum of the solution with a selected parameter at the section Eigenvalue Analysis below.

Linearization of equations of motion

The equations of motion for an axisymmetric body admit equilibrium solutions in which all vectors remain time-invariant in the reference frame of the rotating magnetic field. These equilibria have been confirmed by experimental observations and numerical simulations. Based on this fact, the equations of motion are linearized about the equilibrium point/trajectory defined by the generalized coordinates θ, ϕ, ψ and their first derivatives, using a first-order Taylor expansion.

A Taylor expansion over a vector-valued nonlinear function $G(\mathbf{x})$ can be expressed as follows:

Having a vector of variables $\mathbf{x}$ and the function G as

$$(1)\quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n,\qquad G : \mathbb{R}^n \to \mathbb{R}^m,\qquad G(\mathbf{x}) = \begin{bmatrix} G_1(\mathbf{x}) \\ G_2(\mathbf{x}) \\ \vdots \\ G_m(\mathbf{x}) \end{bmatrix}.$$

The Taylor expansion of G around a point $\mathbf{x_0}$ up to first order is

$$(2)\quad G(\mathbf{x}) \approx G(\mathbf{x_0}) + J_G(\mathbf{x_0})\,(\mathbf{x} - \mathbf{x_0}),$$

where the Jacobian matrix is defined as

$$(3)\quad J_G(\mathbf{x_0}) = \begin{bmatrix} \frac{\partial G_1}{\partial x_1} & \cdots & \frac{\partial G_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial G_m}{\partial x_1} & \cdots & \frac{\partial G_m}{\partial x_n} \end{bmatrix}_{x = x_0}.$$

If $\mathbf{x_0}$ corresponds to an equilibrium $(G(\mathbf{x_0}) = 0)$, this reduces to

$$(4)\quad G(\mathbf{x}) \approx J_G(\mathbf{x_0})\,(\mathbf{x} - \mathbf{x_0}).$$

Applying this formulation to the angular dynamics of the model, the three equations of motion can be expressed as

$$(5)\quad \ddot s = F_s(\theta, \phi, \psi, \dot\theta, \dot\phi, \dot\psi),\quad s \in \{\theta, \phi, \psi\}.$$

The equilibrium condition can be defined as

$$(6)\quad \ddot s = F_s(\theta_0, \phi_0, \psi_0, \dot\theta_0, \dot\phi_0, \dot\psi_0) = 0,\quad s \in \{\theta, \phi, \psi\}.$$

State variables that satisfy this equilibrium can be found as

$$(7)\quad \theta_0 = \text{constant value determined by solving } F_\theta = 0,$$

$$(8)\quad \phi_0 = \omega t + \phi_p,$$

$$(9)\quad \psi_0 = -\kappa\,\omega t,$$

$$(10)\quad \dot\theta_0 = 0,$$

$$(11)\quad \dot\phi_0 = \omega,$$

$$(12)\quad \dot\psi_0 = -\kappa\,\omega,$$

where κ is a coefficient relating the angular velocity term $\dot\psi$ to the angular velocity ω of the rotating field. It is set by the initial conditions and remains constant when damping is zero, but converges to $\cos\theta$ in the presence of damping.

The phase ϕ_p may reach an equilibrium value of π when damping is negligible, while it increases slightly in proportion to the damping.

It should be noted that the phase ϕ_p determines the azimuthal offset between the magnetic moment $\vec{\mathbf{m}}$ and the rotating magnetic field $\vec{\mathbf{B}}$. A value of $\phi_p = \pi$ corresponds to antiparallel alignment of $\vec{\mathbf{m}}$ with $\vec{\mathbf{B}}$, the key phase-lag condition of the dynamics.

In equilibrium condition, the equation $F_\theta = 0$ for an axisymmetric body can be written as

$$(13)\quad -mB\,(\sin\gamma\,\sin\theta_0 - \cos\gamma\,\cos\phi_p\,\cos\theta_0) + \omega^2\,(I_1\kappa - (I_1 - I_3)\,(\kappa - \cos\theta_0))\sin\theta_0 = 0.$$

Under zero damping condition, by setting $\phi_p = \pi$, this simplify to

$$(14)\quad -mB\cos(\gamma - \theta_0) + \omega^2\,(I_1\kappa - (I_1 - I_3)\,(\kappa - \cos\theta_0))\sin\theta_0 = 0,$$

which can be solved numerically. In the case of $\kappa = 1$, this simplify to

$$(15)\quad -mB\cos(\gamma - \theta_0) + \omega^2\,(I_1 + (I_1 - I_3)\,(\cos\theta_0 - 1))\sin\theta_0 = 0,$$

and for $\kappa = 1$ and for a body having isotropic moment of inertia $(I_1 = I_2 = I_3)$, it simplify to

$$(16)\quad -mB\cos\gamma\,\cos\theta_0 + (I_1\omega^2 - mB\sin\gamma)\sin\theta_0 = 0,$$

allowing θ_0 to be derived analytically.

In our model, the generalized coordinates (Euler angles), their velocities and accelerations can be defined as

$$(17)\quad q = \begin{bmatrix} \theta \\ \phi \\ \psi \end{bmatrix},\qquad \dot q = \begin{bmatrix} \dot\theta \\ \dot\phi \\ \dot\psi \end{bmatrix}.$$

We also define corresponding vectors that satisfy the equilibrium state as

$$(18)\quad q_0 = \begin{bmatrix} \theta_0 \\ \phi_0 \\ \psi_0 \end{bmatrix},\qquad \dot q_0 = \begin{bmatrix} \dot\theta_0 \\ \dot\phi_0 \\ \dot\psi_0 \end{bmatrix}.$$

This way the state vector $\mathbf{x}$ and equilibrium state vector $\mathbf{x_0}$ can be defined as:

$$(19)\quad \mathbf{x} = \begin{bmatrix} q \\ \dot q \end{bmatrix} = \begin{bmatrix} \theta \\ \phi \\ \psi \\ \dot\theta \\ \dot\phi \\ \dot\psi \end{bmatrix},\qquad \mathbf{x_0} = \begin{bmatrix} q_0 \\ \dot q_0 \end{bmatrix} = \begin{bmatrix} \theta_0 \\ \phi_0 \\ \psi_0 \\ \dot\theta_0 \\ \dot\phi_0 \\ \dot\psi_0 \end{bmatrix}.$$

The Jacobian $J(\mathbf{x})$ evaluated at the equilibrium point/trajectory $\mathbf{x_0}$ can be expressed as:

$$(20)\quad J(\mathbf{x}) = \left.\frac{\partial\,(F_\theta, F_\phi, F_\psi)}{\partial\mathbf{x}}\right|_{x = x_0} = [\,J(q) \mid J(\dot q)\,],$$

where

$$(21)\quad J(q) = \left.\frac{\partial(F_\theta, F_\phi, F_\psi)}{\partial q}\right|_{x_0},\qquad J(\dot q) = \left.\frac{\partial(F_\theta, F_\phi, F_\psi)}{\partial \dot q}\right|_{x_0}.$$

At the equilibrium point/trajectory, $J(q)$ and $J(\dot q)$ are derived as

$$(22)\quad J(q) = \begin{bmatrix} I_3\omega^2\cos\theta_0 + \frac{mB\,(-\sin\gamma\,\cos\theta_0 - \sin\theta_0\,\cos\gamma\,\cos\phi_p) + \omega^2\,(-I_1 I_3\cos\theta_0 + (-I_1 + I_3)\sin^2(\theta_0) + (I_1\kappa + (-I_1 + I_3)\,(\kappa - \cos\theta_0))\cos\theta_0)}{I_1} & -\frac{mB\sin\phi_p\,\cos\gamma\,\cos\theta_0}{I_1} & 0 \\ \frac{mB\sin\phi_p\,\cos\gamma\,\cos\theta_0}{I_1\sin^2(\theta_0)} & -\frac{mB\cos\gamma\,\cos\phi_p}{I_1\sin\theta_0} & 0 \\ \omega\xi\left(\frac{1}{I_3} - \frac{1}{I_1}\right)\sin\theta_0 - \frac{mB\sin\phi_p\,\cos\gamma}{I_1\sin^2(\theta_0)} & \frac{mB\cos\gamma\,\cos\phi_p\,\cos\theta_0}{I_1\sin\theta_0} & 0 \end{bmatrix},$$

$$(23)\quad J(\dot q) = \begin{bmatrix} -\frac{\xi}{I_1} & \frac{\omega\,(I_3\kappa + 2\,(I_1 - I_3)\cos\theta_0)\sin\theta_0}{I_1} & -\frac{I_3\omega\sin\theta_0}{I_1} \\ \frac{\omega\,(-I_3\kappa + (-2I_1 + I_3)\cos\theta_0)}{I_1\sin\theta_0} & -\frac{\xi}{I_1} & 0 \\ \frac{\omega\,(2I_1 + I_3\,(\kappa\cos\theta_0 - 1) + (-I_1 + I_3)\sin^2(\theta_0))}{I_1\sin\theta_0} & \frac{\xi\,(-I_1 + I_3)\cos\theta_0}{I_1 I_3} & -\frac{\xi}{I_3} \end{bmatrix}.$$

For an isotropic body these Jacobians simplify as

$$(24)\quad J(q) = \begin{bmatrix} \frac{I_1\kappa\omega^2\cos\theta_0 + mB\,(-\sin\gamma\,\cos\theta_0 - \sin\theta_0\,\cos\gamma\,\cos\phi_p)}{I_1} & -\frac{mB\sin\phi_p\,\cos\gamma\,\cos\theta_0}{I_1} & 0 \\ \frac{mB\sin\phi_p\,\cos\gamma\,\cos\theta_0}{I_1\sin^2(\theta_0)} & -\frac{mB\cos\gamma\,\cos\phi_p}{I_1\sin\theta_0} & 0 \\ -\frac{mB\sin\phi_p\,\cos\gamma}{I_1\sin^2(\theta_0)} & \frac{mB\cos\gamma\,\cos\phi_p}{I_1\tan\theta_0} & 0 \end{bmatrix},$$

$$(25)\quad J(\dot q) = \begin{bmatrix} -\frac{\xi}{I_1} & \kappa\omega\sin\theta_0 & -\omega\sin\theta_0 \\ -\frac{\omega\,(\kappa + \cos\theta_0)}{\sin\theta_0} & -\frac{\xi}{I_1} & 0 \\ \frac{\omega\,(\kappa\cos\theta_0 + 1)}{\sin\theta_0} & 0 & -\frac{\xi}{I_1} \end{bmatrix}.$$

Using Jacobians, the equation of motion can be expressed in matrix form as

$$(26)\quad \ddot q = J(q)\,q + J(\dot q)\,\dot q,$$

and in first order state matrix form as

$$(27)\quad \dot{\begin{bmatrix} q \\ \dot q \end{bmatrix}} = \underbrace{\begin{bmatrix} 0_3 & I_3 \\ J(q) & J(\dot q) \end{bmatrix}}_{A} \begin{bmatrix} q \\ \dot q \end{bmatrix},$$

where the term denoted as A is the 6×6 state matrix, 0_3 is a zero matrix and I_3 is an identity matrix in 3×3 form.

Using state vector x, this state matrix representation of equation of motion can be expressed as

$$(28)\quad \dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0_3 & I_3 \\ J(q) & J(\dot q) \end{bmatrix}}_{A} \mathbf{x}.$$

In order to carry out the linearization, the state vector $\mathbf{x}$ is expressed as a perturbation around the equilibrium point/trajectory

$$(29)\quad \mathbf{x} = \mathbf{x_0} + \delta\mathbf{x}.$$

Note that vector $\delta\mathbf{x}$ contains both small variations of coordinates $\delta\mathbf{q}$ and small variations of velocities $\delta\dot{\mathbf{q}}$.

$$(30)\quad \delta\mathbf{x} = \begin{bmatrix} \delta q \\ \delta\dot q \end{bmatrix}.$$

This way, the linearized equations of motion in matrix form using Taylor expansion of F around a point/trajectory x_0 up to first order can be expressed as

$$(31)\quad \dot x = F(\mathbf{x}) \approx F(\mathbf{x_0}) + J(\mathbf{x_0})\,\delta\mathbf{x}.$$

Note that for the steady-rotation equilibrium where second derivatives of coordinates are zero, we have

$$(32)\quad F(\mathbf{x_0}) = \dot{\mathbf{x}}_0 = \begin{bmatrix} \dot q_0 \\ \ddot q_0 \end{bmatrix} = \begin{bmatrix} 0 \\ \dot\phi_0 \\ \dot\psi_0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

so $F(\mathbf{x_0})$ is not the zero vector (because $\dot\phi_0, \dot\psi_0 \neq 0$). Subtract $\dot{\mathbf{x}}_0 = F(\mathbf{x_0})$ from both sides of (29). Since $\delta\dot{\mathbf{x}} = \dot{\mathbf{x}} - \dot{\mathbf{x}}_0$ we get

$$(33)\quad \delta\dot{\mathbf{x}} = \dot{\mathbf{x}} - \dot{\mathbf{x}}_0 = F(\mathbf{x_0}) + J_x(\mathbf{x_0})\,\delta\mathbf{x} - F(\mathbf{x_0}) = J_x(\mathbf{x_0})\,\delta\mathbf{x}.$$

Using the block form of the Jacobian introduced in (27)--(28),

$$(34)\quad J_x(\mathbf{x_0}) = \begin{bmatrix} 0_3 & I_3 \\ J(q) & J(\dot q) \end{bmatrix} = A.$$

In the perturbation form, the state-matrix (A) representation of linearized equations of motion now can be expressed as

$$\boxed{(35)\quad \delta\dot{\mathbf{x}} = A\,\delta\mathbf{x}.}$$

▾ E. Derivation of the Stability Criterion from Eigenvalue Bifurcation

Ucar's 2021 paper introduced a semi-empirical criterion for the stability of the angular motion of an isotropic magnetic body for the simple case where damping is zero and body returns to its initial state after one cycle of the rotating field without making a full turn. In the notation used here, this branch is described by $(\theta = \theta_0)$, $(\phi = \omega t + \phi_p)$, and $(\psi = -\omega t)$, with $(\phi_p = \pi)$, which corresponds to antiparallel alignment of the transverse projection of the body's magnetic moment with the rotating transverse component of the external magnetic field. Under this phase convention, the criterion is defined as

$$(1)\quad \frac{dF}{d\theta} < I\omega^2.$$

where F denotes the right-hand side of the equation of motion about the state variable θ, I is the isotropic moment of inertia, and ω is the angular velocity of the external magnetic field. This condition states that $\frac{dF}{d\theta}$ should be always less than $I\omega^2$ and since $F(\theta)$ has maximum slope at equilibrium angle θ_0 while it crosses zero. The same sinusoidal zero-crossing slope argument was used in the earlier heuristic criterion, although the angular-coordinate notation (θ, φ) in that work was interchanged relative to the notation used here. This criterion can be also written as

$$(2)\quad \left.\frac{dF}{d\theta}\right|_{\theta = \theta_0} < I\omega^2.$$

This criterion was found to be accurate in numerical simulations and was later seen to coincide with the point where the eigenvalues of the linearized system undergo a bifurcation. Therefore, for the synchronized solution under the same constraints, $(\kappa = 1)$ and zero damping, the criterion may be derived from the Jacobian matrices of the corresponding linearized system.

It should be noted that the synchronized equilibrium branch considered here satisfies $(0 < \theta_0 < \pi/2)$, $(0 < \gamma < \pi/2)$, and $(\phi_p = \pi)$, corresponding to antiparallel alignment of the transverse projection of the body's magnetic moment with the rotating transverse component of the external magnetic field. These assumptions define the branch used throughout the derivation; the sign of the zero-crossing slope follows below from the equilibrium condition.

Starting from the Jacobian for an isotropic body Eq.24 and Eq.25 in Section D - Linearization and Eigenvalue Analysis, by applying $(\kappa = 1)$ and zero damping $(\xi = 0)$ constraints, we get:

$$(3)\quad J(q) = \begin{bmatrix} \beta\,(\cos\gamma\,\sin\theta_0 - \cos\theta_0\,\sin\gamma) + \omega^2\cos\theta_0 & 0 & 0 \\ 0 & \frac{\beta\cos\gamma}{\sin\theta_0} & 0 \\ 0 & -\frac{\beta\cos\gamma}{\tan\theta_0} & 0 \end{bmatrix},$$

$$(4)\quad J(\dot q) = \begin{bmatrix} 0 & \omega\sin\theta_0 & -\omega\sin\theta_0 \\ -\frac{\omega\,(1 + \cos\theta_0)}{\sin\theta_0} & 0 & 0 \\ \frac{\omega\,(1 + \cos\theta_0)}{\sin\theta_0} & 0 & 0 \end{bmatrix},$$

where β denotes mB/I. The equilibrium condition of the synchronized solution, $F(\theta_0) = 0$, gives

$$(5)\quad \beta\cos\gamma\cos\theta_0 = (\omega^2 - \beta\sin\gamma)\sin\theta_0.$$

Multiplying the corresponding rearranged form by $\cos\theta_0$, we obtain

$$(6)\quad (\omega^2 - \beta\sin\gamma)\cos\theta_0 = \frac{\beta\cos\gamma\cos^2\theta_0}{\sin\theta_0}.$$

Using Eq. (6), the first element of $J(q)$ becomes

$$(7)\quad \begin{aligned} J_{11} &= (\omega^2 - \beta\sin\gamma)\cos\theta_0 + \beta\sin\theta_0\cos\gamma \\ &= \frac{\beta\cos\gamma\cos^2\theta_0}{\sin\theta_0} + \beta\sin\theta_0\cos\gamma \\ &= \beta\cos\gamma\left(\frac{\cos^2\theta_0}{\sin\theta_0} + \sin\theta_0\right) \\ &= \frac{\beta\cos\gamma}{\sin\theta_0}. \end{aligned}$$

The Jacobians for this solution can therefore be written as

$$(8)\quad J(q)|J(\dot q) = \begin{bmatrix} \frac{\beta\cos\gamma}{\sin\theta_0} & 0 & 0 & 0 & \omega\sin\theta_0 & -\omega\sin\theta_0 \\ 0 & \frac{\beta\cos\gamma}{\sin\theta_0} & 0 & -\frac{\omega\,(1 + \cos\theta_0)}{\sin\theta_0} & 0 & 0 \\ 0 & -\frac{\beta\cos\gamma}{\tan\theta_0} & 0 & \frac{\omega\,(1 + \cos\theta_0)}{\sin\theta_0} & 0 & 0 \end{bmatrix}.$$

Introducing the compact symbols

$$(9)\quad a = \frac{\beta\cos\gamma}{\sin\theta_0},\qquad s = \sin(\theta_0),\qquad c = \cos(\theta_0),$$

the Jacobians take the compact form

$$(10)\quad J(q)|J(\dot q) = \begin{bmatrix} a & 0 & 0 & 0 & s\omega & -s\omega \\ 0 & a & 0 & -\frac{\omega\,(1 + c)}{s} & 0 & 0 \\ 0 & -ac & 0 & \frac{\omega\,(1 + c)}{s} & 0 & 0 \end{bmatrix}.$$

We assume a modal perturbation of the form

$$(11)\quad \delta q(t) = \hat q e^{\lambda t},\qquad \hat q = \begin{bmatrix} \hat\theta \\ \hat\phi \\ \hat\psi \end{bmatrix}.$$

Differentiating Eq. (11) with respect to time gives

$$(12)\quad \delta\dot q(t) = \lambda\,\hat q\,e^{\lambda t},$$

and

$$(13)\quad \delta\ddot q(t) = \lambda^2\,\hat q\,e^{\lambda t}.$$

The linearized perturbation equation is

$$(14)\quad \delta\ddot q = J_q\,\delta q + J_{\dot q}\,\delta\dot q$$

Substituting Eqs. (11)--(13) into Eq. (14), and canceling the common factor $e^{\lambda t}$, gives

$$(15)\quad \lambda^2\hat q = J_q\,\hat q + \lambda J_{\dot q}\,\hat q.$$

Therefore

$$(16)\quad (\lambda^2 I_3 - \lambda\,J_{\dot q} - J_q)\hat q = 0.$$

The modal matrix is therefore

$$(17)\quad M(\lambda) = \lambda^2 I_3 - \lambda\,J_{\dot q} - J_q.$$

Expanding Jacobians matrices above, gives

$$(18)\quad M(\lambda) = \begin{bmatrix} \lambda^2 - a & -\lambda s\omega & \lambda s\omega \\ \frac{\lambda\omega\,(1 + c)}{s} & \lambda^2 - a & 0 \\ -\frac{\lambda\omega\,(1 + c)}{s} & ac & \lambda^2 \end{bmatrix}.$$

A nonzero modal amplitude vector $\hat q \neq 0$ exists only if

$$(19)\quad \det M(\lambda) = \lambda^2\left(a^2 + \lambda^4 + 2\lambda^2\left(-a + \omega^2(1 + c)\right) + a\omega^2\left(-1 + c^2\right)\right) = 0.$$

The determinant shows that the prefactor λ^2 corresponds to two zero eigenvalues. The nontrivial characteristic polynomial is

$$(20)\quad a^2 + \lambda^4 + 2\lambda^2\left(-a + \omega^2(1 + c)\right) + a\omega^2\left(-1 + c^2\right) = 0.$$

The discriminant of this quadratic polynomial in λ^2 is

$$(21)\quad \Delta = -4\omega^2(1 + c)^2\left(a - \omega^2\right).$$

The eigenvalue transition occurs when this discriminant vanishes, which gives

$$(22)\quad a = \omega^2.$$

On the other hand, denoting the derivative of F evaluated at equilibrium as $F_{\theta,0} \equiv \left.\frac{dF}{d\theta}\right|_{\theta = \theta_0}$, we can express $\frac{F_{\theta,0}}{I}$ as

$$(23)\quad \frac{F_{\theta,0}}{I} = \beta\,(\cos\gamma\,\sin\theta_0 - \cos\theta_0\,\sin\gamma) + \omega^2\cos\theta_0.$$

Since the right-hand side of Eq. (23) is identical to the expression for J_{11} before simplification in Eq. (7), we obtain

$$(24)\quad \frac{F_{\theta,0}}{I} = J_{11} = \frac{\beta\cos\gamma}{\sin\theta_0} = a.$$

As the present derivation is restricted to the synchronized branch satisfying $(0 < \theta_0 < \pi/2)$ and $(0 < \gamma < \pi/2)$, Eq. (24) implies $(F_{\theta,0} > 0)$, so the selected equilibrium corresponds to the positive-slope zero crossing of $(F(\theta))$.

Since Eq. (22) gives the eigenvalue transition at $a = \omega^2$, the stable side of the synchronized branch is

$$(25)\quad a < \omega^2.$$

Using Eq. (24), this becomes

$$(26)\quad \frac{F_{\theta,0}}{I} < \omega^2.$$

Therefore,

$$(27)\quad F_{\theta,0} < I\omega^2,$$

or equivalently,

$$(28)\quad \left.\frac{dF}{d\theta}\right|_{\theta = \theta_0} < I\omega^2.$$

Thus, the eigenvalue bifurcation condition recovers the semi-empirical stability criterion.

To obtain the explicit minimum angle, the right-hand side of the θ equation is written as

$$(29)\quad F(\theta) = I\omega^2\sin\theta - mB\,(\cos\gamma\cos\theta + \sin\gamma\sin\theta).$$

Equivalently,

$$(30)\quad F(\theta) = (I\omega^2 - mB\sin\gamma)\sin\theta - mB\cos\gamma\cos\theta.$$

This is a sinusoidal function of θ. Since $F(\theta_0) = 0$, it crosses zero at the equilibrium angle θ_0. The amplitude of this sinusoid is

$$(31)\quad u = \sqrt{(I\omega^2 - mB\sin\gamma)^2 + (mB\cos\gamma)^2}.$$

For the synchronized branch considered here, the equilibrium corresponds to the positive-slope zero crossing. Therefore, the derivative at equilibrium is equal to the amplitude of the sinusoid:

$$(32)\quad F_{\theta,0} = u.$$

Therefore, Eq. (27) becomes

$$(33)\quad u < I\omega^2.$$

Substituting Eq. (31) into Eq. (33) gives

$$(34)\quad \sqrt{(I\omega^2 - mB\sin\gamma)^2 + (mB\cos\gamma)^2} < I\omega^2.$$

Squaring both sides,

$$(35)\quad (I\omega^2 - mB\sin\gamma)^2 + m^2B^2\cos^2\gamma < I^2\omega^4.$$

Expanding,

$$(36)\quad I^2\omega^4 - 2I\omega^2 mB\sin\gamma + m^2B^2\sin^2\gamma + m^2B^2\cos^2\gamma < I^2\omega^4.$$

Using $\sin^2\gamma + \cos^2\gamma = 1$, this becomes

$$(37)\quad I^2\omega^4 - 2I\omega^2 mB\sin\gamma + m^2B^2 < I^2\omega^4.$$

After cancellation,

$$(38)\quad m^2B^2 - 2I\omega^2 mB\sin\gamma < 0.$$

Therefore,

$$(39)\quad 2I\omega^2\sin\gamma > mB.$$

Thus,

$$(40)\quad \boxed{\sin\gamma > \frac{mB}{2I\omega^2}}.$$

The minimum tilt angle is therefore

$$(41)\quad \boxed{\gamma_{\min} = \sin^{-1}\left(\frac{mB}{2I\omega^2}\right)}.$$

The original semi-empirical criterion in the 2021 paper was written using the static and cyclic torque factors, denoted here by τ_S and τ_C. In that notation, the criterion is

$$(42)\quad \tau_S > I\omega^2 - \sqrt{(I\omega^2)^2 - \tau_C^2}.$$

Here τ_S is the maximum torque factor associated with the axial, static field component, and τ_C is the maximum torque factor associated with the radial, cyclic field component. For a field written in terms of the tilt angle γ,

$$(43)\quad \tau_C = mB\cos\gamma,\qquad \tau_S = mB\sin\gamma.$$

Let

$$(44)\quad T = I\omega^2.$$

Then Eq. (42) becomes

$$(45)\quad \tau_S > T - \sqrt{T^2 - \tau_C^2}.$$

Moving terms gives

$$(46)\quad \sqrt{T^2 - \tau_C^2} > T - \tau_S.$$

Squaring both sides,

$$(47)\quad T^2 - \tau_C^2 > T^2 - 2T\tau_S + \tau_S^2.$$

After cancellation,

$$(48)\quad 2T\tau_S > \tau_S^2 + \tau_C^2.$$

Since

$$(49)\quad \tau_S^2 + \tau_C^2 = m^2B^2\sin^2\gamma + m^2B^2\cos^2\gamma = m^2B^2,$$

Eq. (48) becomes

$$(50)\quad 2I\omega^2 mB\sin\gamma > m^2B^2.$$

Dividing by mB gives

$$(51)\quad 2I\omega^2\sin\gamma > mB.$$

Therefore,

$$(52)\quad \boxed{\sin\gamma > \frac{mB}{2I\omega^2}}.$$

Thus, the torque-factor criterion of the 2021 formulation is equivalent to Eq. (40), which was obtained from the eigenvalue bifurcation condition.

Notes:
In the case where the magnetic field $\vec B$ is produced by a magnetic moment $\vec m_R$ rotating about the z-axis, the relation between the elevation angle γ_m of this moment from the xy-plane and the elevation angle of $\vec B$ from the same plane (denoted here as γ_B for clarity) at any point on the z-axis can be found as

$$\boxed{(24)\quad \tan\gamma_B = -2\tan\gamma_m.}$$

This relation can be generalized to any field point $\vec B$ by introducing a local coordinate system whose z-axis is aligned with a vector $\vec r$ joining the dipole moment and the field point. In terms of zenith angles ($\zeta_m = \pi/2 - \gamma_m$, $\zeta_B = \pi/2 - \gamma_B$), one finds

$$\boxed{(25)\quad \cot\zeta_B = -2\cot\zeta_m.}$$

Proof:

Having a point dipole $\mathbf{m}$, its field $\mathbf{B}$ at offset pointed by vector $\mathbf{r}$ reads as

$$(26)\quad \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi r^3}\left(3(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}\right),\qquad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}.$$

By decomposing $\mathbf{m}$ into components parallel and orthogonal to $\mathbf{r}$

$$(27)\quad \mathbf{m} = \mathbf{m}_\parallel + \mathbf{m}_\perp\qquad \mathbf{m}_\parallel = (\mathbf{m}\cdot\hat{\mathbf{r}})\,\hat{\mathbf{r}},\quad \mathbf{m}_\perp = \mathbf{m} - \mathbf{m}_\parallel,$$

and plug in (26), we get

$$(28)\quad \mathbf{B} = \frac{\mu_0}{4\pi r^3}\left(2\mathbf{m}_\parallel - \mathbf{m}_\perp\right).$$

Similarly, by decomposing $\mathbf{B}$ into $\mathbf{B}_\parallel + \mathbf{B}_\perp$ with respect to $\mathbf{r}$, we obtain

$$(29)\quad \mathbf{B}_\parallel = 2\,\frac{\mu_0}{4\pi r^3}\,\mathbf{m}_\parallel,$$

$$(30)\quad \mathbf{B}_\perp = -\frac{\mu_0}{4\pi r^3}\,\mathbf{m}_\perp,$$

Taking component ratios gives

$$(31)\quad \frac{\mathbf{B}_\parallel}{\mathbf{B}_\perp} = -2\,\frac{\mathbf{m}_\parallel}{\mathbf{m}_\perp} \implies \cot\zeta_B = -2\cot\zeta_m,$$

where ζ_B and ζ_m are the angles between the corresponding vectors and vector $\mathbf{r}$. The negative sign arises from Eq. (30), which shows that the orthogonal components of these vectors are antiparallel.